

The Log Normal Distribution

Description

Density, distribution function, quantile function and random generation for the log normal distribution whose logarithm has mean equal to `meanlog` and standard deviation equal to `sdlog`.

Usage

```
dlnorm(x, meanlog = 0, sdlog = 1, log = FALSE)
plnorm(q, meanlog = 0, sdlog = 1, lower.tail = TRUE, log.p = FALSE)
qlnorm(p, meanlog = 0, sdlog = 1, lower.tail = TRUE, log.p = FALSE)
rlnorm(n, meanlog = 0, sdlog = 1)
```

Arguments

- `x, q` vector of quantiles.
- `p` vector of probabilities.
- `n` number of observations. If `length(n) > 1`, the length is taken to be the number required.
- `meanlog, sdlog` mean and standard deviation of the distribution on the log scale with default values of 0 and 1 respectively.
- `log, log.p` logical; if TRUE, probabilities `p` are given as `log(p)`.
- `lower.tail` logical; if TRUE (default), probabilities are $P[X \leq x]$, otherwise, $P[X > x]$.

Details

The log normal distribution has density

$$f(x) = \frac{1}{\sqrt{2\pi\sigma x}} e^{-(\log(x)-\mu)^2/2\sigma^2}$$

where μ and σ are the mean and standard deviation of the logarithm. The mean is $E(X) = \exp(\mu + 1/2\sigma^2)$, the median is $med(X) = \exp(\mu)$, and the variance $Var(X) = \exp(2\mu + \sigma^2)(\exp(\sigma^2) - 1)$ and hence the coefficient of variation is $\sqrt{\exp(\sigma^2) - 1}$ which is approximately σ when that is small (e.g., $\sigma < 1/2$).

Value

`dlnorm` gives the density, `plnorm` gives the distribution function, `qlnorm` gives the quantile function, and `rlnorm` generates random deviates.

The length of the result is determined by `n` for `rlnorm`, and is the maximum of the lengths of the numerical arguments for the other functions.

The numerical arguments other than `n` are recycled to the length of the result. Only the first elements of the logical arguments are used.

Note

The cumulative hazard $H(t) = -\log(1 - F(t))$ is `-plnorm(t, r, lower = FALSE, log = TRUE)`.

Source

`dlnorm` is calculated from the definition (in ‘Details’). `[pqr]lnorm` are based on the relationship to the normal.

Consequently, they model a single point mass at `exp(meanlog)` for the boundary case `sdlog = 0`.

References

Becker, R. A., Chambers, J. M. and Wilks, A. R. (1988) *The New S Language*. Wadsworth & Brooks/Cole.

Johnson, N. L., Kotz, S. and Balakrishnan, N. (1995) *Continuous Univariate Distributions*, volume 1, chapter 14. Wiley, New York.

See Also

[Distributions](#) for other standard distributions, including [dnorm](#) for the normal distribution.

Examples

```
dlnorm(1) == dnorm(0)
```

[Package *stats* version 4.3.0 [Index](#)]